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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A' Grade

SEMESTER END EXAMINATIONS FEBRUARY - MARCH 2021

Program	:	**B.E. : Computer Science and Engineering**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**CS32**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:
- Answer one full question from each unit

UNIT- I

1. a) Explain dynamic memory allocation with suitable examples. CO1 (06)

   b) Compute the failure functions for the following patterns using KMP algorithm. CO1 (06)
      i) aabaaabab
      ii) abbabbaba

   c) Write necessary C functions to add two polynomials represented using an array of structures. CO1 (08)

2. a) Find the simple transpose of the following sparse matrix. CO1 (06)

Row	Col	Value
6	7	9
0	2	14
0	5	7
1	3	9
2	0	5
2	4	12
3	1	11
4	2	99
4	4	16
5	6	15

   b) Write an ADT for a sparse matrix and include necessary operations. CO1 (08)

   c) Write a recursive C function to find the sum of array elements. CO1 (06)

UNIT – II

3. a) Apply the knowledge of Stack to implement infix to postfix conversion. CO2 (05)

   b) List applications of stacks. Using stack write an algorithm to determine if a given string is palindrome and print suitable message as output. CO2 (10)

   c) What are the advantages of circular queue. CO2 (05)

4. a) A circular queue has a size of 5 and has 3 elements 10,40 and 20, where F=2 and R=4. After inserting 50 and 60,what is value of F and R. Trying to insert 30 at this stage what will happen? Delete 2 elements from the queue and insert 100. Show the sequences of steps with necessary diagrams with the value of F and R. CO2 (06)

   b) Write a C functions which demonstrate queue operations. CO2 (08)

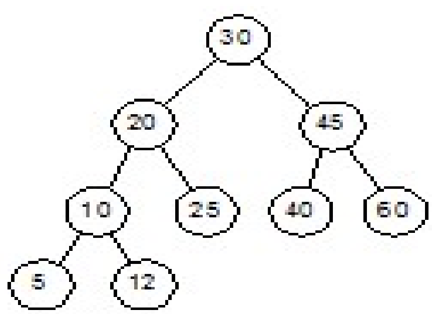
   c) Write an algorithm to convert a given infix expression into prefix expression. CO2 (06)

UNIT – III

5.	a)	Write a C function to delete a node in a doubly circular linked list. Note: Pass the following arguments only - the last nodes address, the data of the node to be deleted.	CO3	(08)
	b)	Write a C function to add two polynomials represented as circular linked list with header nodes. Store the resultant polynomial in a circular linked list.	CO3	(12)
6.	a)	Write a C function that returns TRUE if a word or phrase is present in a linked list of words, and FALSE if it is not .(Hint : Each of the word or phrase is supposed to be stored in doubly linked list nodes.)	CO3	(08)
	b)	Write a C function to find union of two doubly linked lists by concatenating them. The duplicated contents have to be removed in the resultant linked list.	CO3	(08)
	c)	Show the advantages of linked lists over arrays.	CO3	(04)

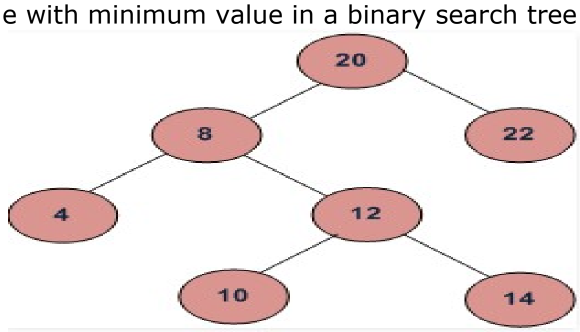
UNIT – IV

7.	a)	Describe the left child-right sibling representation of a tree with suitable example.	CO4	(08)
	b)	Construct the binary tree stepwise from its preorder (ABCDFEG) and inorder(BAFDCEG) traversals.	CO4	(06)
	c)	Write a C function swap_tree that takes a binary tree and swaps the left and right children of every node.	CO4	(06)
8.	a)	Define with an example each of the following: i) Binary tree ii) Full binary tree iii) Complete binary tree.	CO4	(06)
	b)	Describe satisfiability problem. Justify how it can be solved using binary tree with a suitable example.	CO4	(08)
	c)	Write the Pre-Order, In-Order and Post-Order traversals of the following binary tree.	CO4	(06)

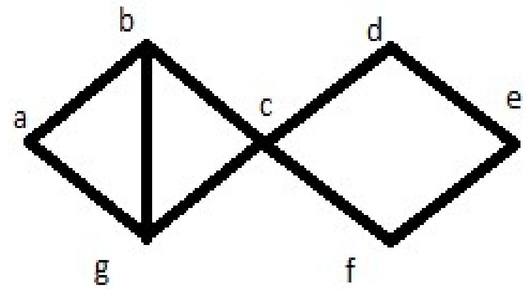


UNIT – V

9.	a)	Find the node with minimum value in a binary search tree.	CO5	(06)



b) Construct the spanning tree using both DFS and BFS. CO5 (06)



c) Compare and contrast the adjacency matrix with adjacency lists representation of graph. CO5 (08)

10. a) Explain the following terms with respect to graphs. Graph, sub-graph, path, node and edge. CO5 (05)

b) Consider we have following key values: CO5 (10)
27,35,22,18,23,38,47,10,14,11
Write out the AVL tree after each value is inserted into the tree.

c) Write C function to find the in-degree and out-degree of each node of directed graph. CO5 (05)

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